Concept sheet: Fooled by Randomness, Part I, Ch. 3 — red-sheet

Answer in your head, then check. Page references are printed book pages.

Math as a tool (pp. 40–45)

1. [key move]
Taleb says this simulation approach is far more a way of thinking than a computational method; mathematics is principally a tool to meditate, rather than to compute.

2. [contrast]
Taleb's ideal: the Europlayboy's realism without the shallowness, combined with the mathematician's intuition without the excessive abstraction.

3. [definition]
A sample path is a sequence of scenarios along the course of time, unlike MBA-style scenario analysis, which looks only at the end outcome.

4. [predict]
For an investor, the path view cares less about his worth in a year than about the heart-wrenching ride along the way.

5. [definition]
A stochastic process studies the evolution of successive random events; the key about a process is that it has time in it.

6. [in words]
A computer-simulated roulette wheel beats one built by a carpenter, whose wheel may favor some numbers owing to a slant in its build (biases).

7. [key move]
Monte Carlo method: set conditions believed to resemble those that prevail in reality, launch thousands or millions of paths, and look at the prevalent characteristics of their features.

8. [contrast]
“True” mathematicians dislike Monte Carlo, calling it brute force that robs mathematics of finesse and elegance.

9. [trap]
Computing Pi by shooting random bullets at a circle in a square is not an efficient use of a computer, but it can give more intuition than lines of equations.

10. [contrast]
A “native” mathematician wants to improve mathematics via theorems and proofs; Taleb, a quant, cares only about the application.

11. [fact]
Taleb could concentrate on an equation only when motivated by a real problem (with a modicum of greed).

12. [fact]
Einstein's 1905 work on Brownian movement can be the backbone of the random walk theories used in financial modeling.

13. [trap]
Keynes, author of a probability treatise, blew up trading his account: people's understanding of probability does not translate into their behavior.

Attic simulations (pp. 45–47)

14. [key move]
Dispersion between simulated results indicates the degree of resistance to randomness.

15. [fact]
The real dividend of the computer revolution for Taleb: fast processors able to generate a million sample paths per minute.

16. [predict]
In his simulated market with idiotic bulls, impetuous bears and cautious traders, bears dropped out in the rally and bulls got slaughtered when the music stopped.

17. [trap]
The one group with remarkable staying power were option buyers, because they could buy insurance against blowup.

18. [definition]
"Summing under histories" (from Feynman): Taleb can no longer visualize a realized outcome without reference to the non-realized ones.

History (pp. 47–51)

19. [trap]
By "history" Taleb means anecdotes, not grand-scale historicism that tries to uncover laws in the evolution of history.

20. [key move]
Historicism is pseudoscientific because it draws theories from past events without allowing that they may have arisen from randomness, and no controlled experiment can verify the claims.

21. [in words]
Two ways Taleb learns from history: from the past, by reading the elders and from the future, via his Monte Carlo toy.

22. [predict]
Children stop touching a burning stove only when they are themselves burned; Taleb infers that our genes do not favor transfers of experience.

23. [fact]
The colleagues who blew up in 1987, 1990 (Japan), 1994, 1998 (Russia) and 2000 (Nasdaq) all claimed "these times are different"; Taleb has yet to meet a denigrator of history who has not blown up.

24. [definition]
Historical determinism: we think we would know when history is made, e.g. that witnesses of the 1929 crash knew they were living an acute historical event.

25. [key move]
Though history cannot be experimented on, in the medium to long run it delivers most of the possible scenarios, burying the bad guy.

26. [definition]
Mathematicians call the idea that bad trades catch up with you ergodicity: very long sample paths end up resembling each other.

27. [predict]
A lottery-winning janitor living 1000 years cannot be expected to win more lotteries; the lucky fool converges to the state of a less-lucky idiot.

Information (pp. 51–56)

28. [contrast]
Analogy: noise vs information corresponds to journalism vs history.

29. [key move]
An idea that has survived many cycles shows relative fitness: some noise has been filtered out.

30. [key move]
Progress means some new information is better than past information, not that the average of new information supplants the past; so, when in doubt, reject the new idea.

31. [trap]
"New things" look good because we see and count only the winners, not the losers.

32. [key move]
Opportunity cost of missing a “new new thing” is minuscule next to the toxicity of the garbage one must wade through.

33. [key move]
The problem with information is not that it is useless but that it is toxic; the media is paid to get your attention.

34. [fact]
Shiller's 1981 paper: stock prices are far too volatile relative to dividends, the fundamentals they are supposed to reflect.

35. [trap]
Merton attacked Shiller on methodological grounds only, yet later founded a hedge fund meant to exploit market inefficiencies, implying he agreed with Shiller.

36. [key move]
Gerontocracy: choosing traders by cumulative years of experience, not absolute success (conditional on not having blown up) gave a significant advantage in simulation.

37. [key move]
Under regime switching it is unclear who is actually the fittest; survivors are the oldest, who have been exposed longer to the rare event.

The dentist (pp. 56–59)

38. [in words]
The wise man listens to meaning; the fool only gets the noise.

39. [fact]
The dentist's expected return is 15% over Treasury bills with 10% volatility: about 68 of 100 paths fall between 5% and 25%, and 95 between −5% and 35%.

40. [fact]
Probability of making money: over one year 93%; over one second 50.02%.

41. [fact]
Table 3.1: over a quarter 77%; over a month 67%; over a day 54%.

42. [key move]
A loss hurts the dentist more than an equal gain pleases him, so at high frequency he runs a large emotional deficit.

43. [fact]
Noise per one part of performance: over one year 0.7; over one second 1796.

44. [key move]
Over a short time increment one observes the variability (variance) of the portfolio, not its returns.

45. [trap]
Same logic: the news, at high frequency, is full of noise, while history, at low frequency, is largely stripped of it.

46. [fact]
Taleb copes by having no access to information except in rare circumstances; if an event is important enough, it will find its way to his ears.

47. [fact]
People who look too closely at randomness burn out; some behavioral economists put the pang of a loss at up to 2.5 times a gain's pleasure.

Vocabulary (standard terms) — term → meaning

Monte Carlo simulation	replaying a random process many times to see the range of outcomes
Sample path	one realized run of a random process through time
Stochastic process	a random process that evolves over time
Random walk	path where each step is random and independent of the past
Volatility	size of typical fluctuation, usually standard deviation per year
Normal distribution	bell curve: about 68% within ±1σ, 95% within ±2σ
Uniform distribution	every point or value equally likely
Signal-to-noise ratio	meaningful movement relative to random fluctuation
Efficient market hypothesis	prices already reflect all available information
Loss aversion	a loss hurts more than an equal gain pleases
Regime switching	market behavior changes between distinct states (boom, bust)

Symbols (none appear in pp. 40–59; common in the field) — symbol → name

μ	mean (expected return)
σ	standard deviation (volatility)
σ√T	volatility over horizon T grows with the square root of time
N(μ, σ²)	normal distribution with mean μ and variance σ^2
E[X]	expected value

Where it shows up — *my additions, not in the book*

• Trading bot: one backtest is one sample path. Resample or simulate trade sequences to see the spread of drawdowns before trusting it.
• LP project: simulate many price paths to see fee income against rare large losses.
• Options study: for each position, ask who pays for the insurance and who is paid for selling it.
• X / Pixiv analytics: hourly engagement is mostly noise. Read it weekly, and judge by process, not by the last post.
• Same thread: Ch. 5 regime switching and survival (the book's forward reference).

Concept sheet: Fooled by Randomness, Part I, Ch. 3 — blanked

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Review key

1. **far more a way of thinking than a computational method / a tool to meditate, rather than to compute** — p.41.
2. **without the shallowness / without the excessive abstraction** — p.41.
3. **a sequence of scenarios along the course of time / the end outcome** — pp.41–42.
4. **his worth in a year / the heart-wrenching ride along the way** — p.42.
5. **the evolution of successive random events / it has time in it** — p.42. Taleb: “the mathematics of history”.
6. **favor some numbers owing to a slant in its build (biases)** — pp.42–43.
7. **believed to resemble those that prevail in reality / the prevalent characteristics of their features** — p.43.
8. **brute force** — p.43.
9. **not an efficient use of a computer / more intuition than lines of equations** — pp.43–44. The book's “multiple of Pi, infinite precision” wording is loose, so it is not blanked.
10. **to improve mathematics via theorems and proofs / the application** — p.44.
11. **a real problem (with a modicum of greed)** — p.44. Same as compulsive gamblers who learn card counting.
12. **Brownian movement / the random walk theories used in financial modeling** — p.45.
13. **blew up trading his account / does not translate into their behavior** — p.45.
14. **the degree of resistance to randomness** — p.45.
15. **fast processors able to generate a million sample paths per minute** — p.46.
16. **dropped out in the rally / got slaughtered when the music stopped** — pp.46–47.
17. **option buyers / buy insurance against blowup** — p.47.
18. **a realized outcome without reference to the non-realized ones** — p.47.
19. **anecdotes / grand-scale historicism** — p.48.
20. **draws theories from past events without allowing that they may have arisen from randomness / no controlled experiment can verify the claims** — p.48.
21. **from the past, by reading the elders / from the future, via his Monte Carlo toy** — p.48.
22. **they are themselves burned / do not favor transfers of experience** — p.48.
23. **“these times are different” / has not blown up** — p.49.
24. **we would know when history is made** — p.49.
25. **delivers most of the possible scenarios, burying the bad guy** — p.50.
26. **ergodicity / end up resembling each other** — p.50. “Roughly, under certain conditions.”
27. **cannot be expected to win more lotteries / converges to the state of a less-lucky idiot** — pp.50–51. Each reverts to his long-term properties.
28. **journalism vs history** — p.51.
29. **relative fitness / has been filtered out** — p.52.
30. **not that the average of new information supplants the past / reject the new idea** — p.52. Taleb: “systematically… always”.
31. **we see and count only the winners, not the losers** — p.52.
32. **minuscule / the toxicity of the garbage one must wade through** — p.52.
33. **toxic / is paid to get your attention** — pp.52–53.
34. **far too volatile relative to dividends** — p.53.
35. **on methodological grounds only / a hedge fund meant to exploit market inefficiencies** — p.54.
36. **cumulative years of experience, not absolute success / a significant advantage** — p.55.
37. **it is unclear who is actually the fittest / have been exposed longer to the rare event** — p.55.
38. **meaning / the noise** — p.56. Philostratus, via Cavafy.
39. **5% and 25% / −5% and 35%** — p.56. ±1 and ±2 standard deviations; checked by code.
40. **93% / 50.02%** — p.57. Table 3.1.
41. **77% / 67% / 54%** — p.57. Checked by code (77.3, 66.8, 53.8).
42. **more than an equal gain pleases him / a large emotional deficit** — pp.57–58.

43. **0.7 / 1796** — p.58. Checked by code (0.67 and 1796).
44. **the variability (variance) of the portfolio, not its returns** — p.58.
45. **is full of noise / is largely stripped of it** — p.59.
46. **no access to information except in rare circumstances / it will find its way to his ears** — pp.58–59.
47. **burn out / 2.5 times** — p.59.